Himani Kamboj

Haryana, India (Open to relocate to Germany)

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Summary

Embedded Hardware Design Engineer with 5+ years of experience in electronic system design, PCB layout, and hardware validation in surveillance and imaging systems. Strong expertise in schematic design, multi-layer PCB development, and power electronics. Proven experience in impedance considerations, component selection, and board bring-up. Skilled in working with cross-functional teams to deliver reliable embedded products.

Core Skills

- **Hardware Design:** Electronic Circuit Design, Schematic Capture, PCB Layout, Multi-layer PCB, Signal Integrity, EMI/EMC
- **Design Tools:** Altium Designer, Eagle (Autodesk), Proteus
- **Embedded Systems:** Microcontrollers (STM32, Atmega, Attiny), Embedded C
- **Programming:** Embedded C
- **Communication Protocols:** UART, SPI, I2C, MIPI
- **Testing & Validation:** Board Bring-up, Hardware Debugging, Validation Testing
- **Domain Expertise:** Imaging Systems, Surveillance Camera, IP Camera

Experience

Xpia Electronics Pvt. Ltd. **June 2025 – Present**

Lead Design Engineer *Delhi, India*

- Designed end-to-end hardware architecture for IP-based CCTV camera systems using Augentix SoC, enabling stable real-time video processing and network integration, improving system reliability 20%.
- Developed multi-layer PCB layouts and detailed schematics by applying signal integrity and EMI/EMC principles, ensuring compliance with high-speed design requirements.
- Integrated Ethernet PHY and image sensors (CMOS - 2MP/4MP/5MP) by implementing MIPI and UART interfaces, enabling efficient data transmission and processing pipelines.
- Defined and executed hardware validation strategies to verify system performance against design specifications and operational requirements.
- Optimized power distribution circuits and supported component selection to enhance system stability and meet production and reliability standards, reducing power losses 15%.

Enord Pvt. Ltd. **November 2024 – June 2025**

Design Engineer *Delhi, India*

- Designed expansion PCB board for different payloads, enabling reliable real-time flight control operations.
- Integrated sensors (IMU, GPS, barometer, IR) and communication protocols (UART, SPI, I2C) to support accurate navigation and control systems, improving signal reliability 20%.
- Designed and implemented UAV avionics schematics and PCB layouts using STM32F7 microcontrollers, enabling stable real-time flight control operations.
- Integrated and validated embedded control systems by interfacing sensors and communication protocols, ensuring reliable data acquisition and flight system responsiveness.
- Supported component selection and qualification aligned with defense-grade reliability standards, ensuring system robustness in critical environments.

TAK Technologies Pvt. Ltd. **May 2022 – November 2024**

Design Engineer *Noida, India*

- Designed hardware architecture and PCB layouts for night vision monocular systems using Atmega microcontrollers, enabling reliable low-light surveillance capabilities.
- Integrated image intensifier tubes, IR lasers, and sensors to enhance targeting accuracy and system performance in low-visibility environments, improving detection efficiency 20%.
- Delivered end-to-end product integration across mechanical, optical, and electronic components, ensuring system functionality and field readiness.
- Designed compact and efficient circuits for holographic weapon sight systems using Attiny microcontrollers, improving system ruggedness and usability.

Accuster Technologies Pvt. Ltd. **October 2020 – May 2022**

Electronics Engineer, R&D *Delhi, India*

- Designed hardware architecture and PCB layouts for automated microscope systems, enabling conversion from manual to automated operation, improving process efficiency 30%.
- Integrated sensors and communication interfaces to support accurate blood cell counting and system functionality.
- Performed hardware validation and debugging to ensure stable performance and reliability in medical applications.

Training

- **PCB Design using Altium Designer** – Sofcon Pvt. Ltd., 45-day intensive training in schematic capture and multi-layer PCB design
- **Arduino Hardware and PCB Design** – 3-month hands-on training focused on embedded systems development and circuit design
- **Industrial Training in Embedded Systems (Arduino)** – 6-week program covering microcontroller programming and hardware interfacing
- **Thermal Imaging Camera Development Workshop** – CDAC, 4-day workshop covering thermal imaging system development, sensor calibration, thermal optimization, and noise reduction techniques

Education

Haryana Engineering College, Kurukshetra University **2016 – 2020**

Bachelor of Technology in Electronics and Communication Engineering *Haryana, India*

Key Achievements

- Served as Team Coordinator as a Design Engineer by facilitating seamless integration of cross-functional technical teams for efficient project execution.
- Improved execution efficiency by systematically maintaining and sharing critical engineering data with journal engineers, enabling smooth project planning and implementation.
- Participated in a college-level hackathon and contributed to the development of an OFDM (Orthogonal Frequency Division Multiplexing) based communication project.

Languages

- English – Full Professional Proficiency (C1)
- German – Intermediate Proficiency (B1)